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13 SEPTEMBER 1978
SUPERSEDING
MIL-L-9965(USAF)
30 January 1963

MILITARY SPECIFICATION

LIGHT, WARNING, VEHICULAR, FLASHING

This specification is approved for use by
HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016.
Department of the Air Force, and is available for use by all
Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers 12 volt and 24 volt nonrotating, flashing, vehicular mounting, warning lights.

1.2 Classification. Warning lights shall be of the following types, as specified (see 6.2):

Type I For 12 volt DC operation
Type II For 24 volt DC operation

2. APPLICABLE DOCUMENTS

* 2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

FF-B-588	Bolt, Toggle: And Expansion Sleeve, Screw
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed And Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Edged And Components
PPP-B-676	Boxes, Setup
PPP-T-60	Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6220

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Military

MIL-P-116	Preservation-Packaging, Methods Of
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, And Sheet, Overwrap; Water-Vaporproof Or Waterproof Flexible
MIL-C-25050	Colors, Aeronautical Lights And Lighting Equipment, General Requirements For

STANDARDS

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification, to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J575 - Tests for Motor Vehicle Lighting Devices and Components

(Copies of SAE publications may be obtained from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale PA 15096.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Tariff Order Section, 1616 P Street, N.W., Washington DC 20036.)

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3. REQUIREMENTS

3.1 Preproduction. Preproduction samples shall be furnished for inspection and testing as specified in 4.3.

3.2 Materials. Material not otherwise specified herein shall be of such quality as to be free from any defects that might adversely affect performance. Materials having a tendency to warp or to absorb moisture shall not be used.

3.3 Design and construction requirements. The light shall be designed to mount on a vehicular roof top by means of 3 toggle bolts through the base plate. Mounting holes shall be 3 holes of 0.265 inches in diameter, equally spaced on a bolt circle diameter of 6.250 inches. The lamp shall have a dome having a vertical axis, base, base gasket or mounting pad, switch, lamp, circuit interrupter and suitable features to allow 12 volt or 24 volt adaptation.

* 3.3.1 Dome. The dome shall be molded plexiglass, lucite or other plasticized material. Glass shall not be furnished. Color shall be aviation red, blue or yellow, in accordance with MIL-C-25050. It shall provide visibility of the light for 360 degrees about its vertical axis from its horizontal base to 90 degrees above.

3.3.2 Base. A base of nonferrous metal suitable for support of the light during operations described in 3.4 shall be provided. The base shall be of the receding type for mounting on flat or curved surfaces or shall have a skirt of configuration and metal content to allow fitting to vehicle roof contours. The base shall have a base plate with mounting holes as required by 3.3.

3.3.3 Base gasket. A gasket or mounting pad of rubber, rubber composite, or other suitable material shall be provided as a seal between the vehicle roof and light base. It shall be flexible and of sufficient length or size to fit any roof contour on which the light may be mounted. It shall meet the requirements of 3.4.

* 3.3.4 Mounting bolts. Three 1/4 inch diameter toggle bolts in accordance with FF-B-588, Type I or II, Class A or B, Style 1 or equivalent shall be furnished to anchor the light to vehicle roof tops.

* 3.3.5 Lamp. A lamp or lamps when tested as specified in 4.3.4 shall be provided to furnish illumination of the candlepower required in Table I. The manufacturer's standard lamp shall be furnished provided it meets requirements of this specification.

* TABLE I. Photometric Minimum Candlepower Requirements.

TEST ANGLE DEGREES	MINIMUM CANDLEPOWER			
	WHITE	YELLOW	RED	BLUE
5 up to 5 down	278	125	50	8
2.5 up to 2.5 down	1110	500	200	30

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3.3.6 Circuit interrupter. An off and on switch and an automatic flasher device shall be furnished to provide a minimum of 80 and a maximum of 90 circuit interruptions or light flashes per minute.

3.3.7 Electrical. The lights shall be for use in 12 to 16 or 24 to 28 volt circuits as specified (see 6.2). The design shall allow conversion from one circuit to the other by changing the lamp and flasher unit or other small components.

3.3.7.1 Wire. Wire shall be No. 14 AWG stranded insulated wire not less than 15 feet in length. No connector is required on the free end.

* 3.3.7.2 Radio interference suppression. The light shall be suppressed in accordance with MIL-STD-461 as a nontactical vehicle requirement.

3.4 Performance. The light shall be capable of operating satisfactorily under the following conditions:

a. Temperatures ranging from minus 65° Fahrenheit (F) to plus 160°F.

b. Exposure to salt-sea atmosphere.

c. Vibration incident to service use.

d. Fungus growth as encountered in humid climates.

e. Rainfall as encountered in any locale.

f. Sand and dust conditions as may be encountered in arid locations.

3.5 Installation instructions. Each warning light shall be furnished with complete instructions covering the installation of the light including a template. A wiring diagram showing all electrical connections to be made shall be a part of these instructions.

* 3.6 Color. The color of the dome shall conform to MIL-C-25050 and shall be aviation red, aviation blue or aviation yellow, as specified in 6.2.

3.7 Workmanship. Workmanship shall be such as to assure warning lights to be free of loose components; also, free of rust, scratches, burrs, sharp edges, chips and unnecessary oil or foreign materials.

3.8 Identification marking. Equipment, assemblies, and parts shall be in accordance with MIL-STD-130.

3.9 Finishing and painting. The equipment shall be finished and painted or plated in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requiremenets specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the

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inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to preceived requirements.

4.2 Classification of tests. The inspection and testing of the light shall be classified as follows:

a. Preproduction tests.

b. Acceptance tests.

4.3 Preproduction inspection and test. A sample of each type of light (see 1.2) the supplier proposes to furnish under this contract shall be inspected and tested as follows:

4.3.1 Examination of product. The preproduction sample shall be thoroughly examined to assure conformance to all requirements of this specification. Characteristics not covered by this specification shall be examined to assure conformance to the manufacturer's specification.

4.3.2 Environmental tests. Unless otherwise specified, the light with lamp installed shall be subjected to the following tests:

4.3.2.1 High temperature. The light shall be placed in the test chamber having a temperature of 160°F. The light shall be exposed to that temperature for a period of 24 hours. A relative humidity of not more than 15 per cent shall be maintained in the test chamber throughout the exposure period. At the conclusion of the exposure period and while still at the test temperature, the light shall be subjected to and shall meet the operational requirements of this specification. Light temperature shall then be returned to that of standard atmospheric conditions and the light shall be again subjected to and shall meet the operational requirements of this specification.

4.3.2.2 Low temperature. The light shall be placed in a test chamber having a temperature of minus 65°F. The light shall be exposed to that temperature for a period of 24 hours. At the conclusion of the exposure period and while still at the test temperature, the light shall be subjected to and shall meet the operational requirements of this specification. The light temperature shall then be returned to that of standard atmospheric conditions and the light again shall be subjected to and shall meet the operational requirements of this specification.

* 4.3.2.3 Vibration. The light shall be subjected to vibration in accordance with vibration test SAE J575. Upon completion of the test the light shall be subjected to and shall meet the operational requirements of this specification.

* 4.3.2.4 Moisture. The light shall be subjected to moisture in accordance with moisture test of SAE J575. Upon completion of this test, the light shall be subjected to and shall meet the operational requirements of this specification.

4.3.2.5 Dust. The light shall be subjected to dust in accordance with dust test of SAE J575. Upon completion of this test, the light shall be subjected to and shall meet the operational requirements of this specification.

* 4.3.2.6 Corrosion. The light shall be subjected to corrosion in accordance with corrosion test of SAE J575. Upon completion of this test, the light shall be subjected to and shall meet the operational requirements of this specification.

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* 4.3.3 Radio interference suppression test. The light shall be tested in accordance with Class III, specified in MIL-STD-461.

4.3.4 Performance test. To determine conformance with 3.3.5 the lamp assembly shall be mounted as in intended operation, the lamps alone electrically connected to a 14 ±0.5 volt or a 28 ±0.5 volt power source, as applicable. Photometric readings for both lamps shall be taken at a 20 foot distance, on the vertical axis. Through the range of minus 5 degrees through plus 30 degrees of the vertical axis, a total of 15 readings shall be recorded in increments of 2-1/2 degrees. The beam candlepower shall be considered to be the value computed from the maximum reading taken. A Wratten Number 102 filter shall be utilized in conjunction with a barrier layer type cell as a receptor for purposes of this test. The warning light shall then be allowed to operate for 150 hours after which a photometric reading shall be obtained using a lamp previously tested. If the lamp fails, the photometric test shall be repeated using a second lamp.

4.3.5 Examination of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to insure compliance with the requirements of Section 5 of this specification and the contract.

4.3.6 Failure. Failure of a sample to pass any of the examination or tests specified herein may be cause for the Government to refuse to conduct additional inspection until the faults revealed by the examination or tests have been corrected.

4.4 Acceptance inspection.

4.4.1 Sampling for warning lights.

4.4.1.1 Lot formation. A lot shall consist of all warning lights of one type, and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.2 Sampling for acceptance testing. Samples for acceptance testing shall be selected in accordance with inspection level L8 of Standard MIL-STD-105. Each sample selected shall be tested for compliance with 3.3.6.

4.4.2 Examination inspection for warning lights.

4.4.2.1 Acceptable quality level. Each sample selected in accordance with 4.4.1.2 shall be examined for conformance to the following accertable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

* 4.4.2.2 Examination. Examination shall be conducted as specified in Table II. Any warning light in the sample containing one or more defects shall be rejected, and if the number of defective warning lights in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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TABLE II. Classification of defects in warning lights in accordance with MIL-STD-105.

Categories	Defects
Critical:	None defined
Major:	
101	Defective materials used (see 3.2).
102	Warning light does not conform to spec (see 3.3).
103	Warning light not properly marked (see 3.8).
104	Improper color (see 3.6).
Minor:	
201	Poor workmanship (see 3.7), not affecting serviceability.

5. PACKAGING

5.1 General. Preservation, packaging, and packing pertains only to direct purchases by the Government. When the light is furnished as part of a vehicle or other end item, these provisions do not apply.

5.2 Preservation and packaging.

5.2.1 Level A. Each light shall be preserved in accordance with MIL-P-116, method III, and individually packaged in containers conforming to PPP-B-636.

5.2.2 Level C. Lights shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.3 Intermediate packaging. Only identical items shall be included in an intermediate package. For items preserved and packaged in accordance with 5.2.1, the requirements for intermediate packaging shall be at the option of the contractor. Lights preserved and packaged in accordance with 5.2.2 shall be packaged in containers conforming to PPP-B-566, PPP-B-636, PPP-B-676, or PPP-B-665. Unless otherwise specified by the procuring activity, the quantity of unit packages to be included in each intermediate package shall be at the option of the contractor and as governed by the limitations of the container being used.

5.4 Packing.

5.4.1 Level A. Lights preserved and packaged as specified in 5.2.1 shall be packed in exterior-type shipping containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636, or MIL-B-10377. Insofar as practicable, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance

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with the appendix thereto. The case liner shall not be required when the unit, intermediate, or exterior container conforms to PPP-B-636, and is sealed at all joints and seams, including the manufacturer s joint with tape conforming to PPP-T-60.

5.4.2 Level B. Lights preserved and packaged as specified in 5.2.1 and 5.3 shall be packed in domestic-type exterior containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. Insofar as practicable, exterior containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall conform to the special requirements table of PPP-B-636, as applicable.

5.4.3 Level C. Packages which require overpacking shall be packed to insure carrier acceptance and safe delivery to destination in compliance with Official Classification Committee, Uniform Freight Classification Rules, or American Trucking Association, National Motor Freight Classification Rules or with other carrier rules and regulations applicable to the mode of transportation.

5.5 Physical protection. Cushioning, blocking, bracing, and bolting as required shall be in accordance with MIL-STD-1186 except that waterproofing requirements for cushioning materials and containers shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5.6 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lights covered by this specification are intended for use on vehicles having 12-volt or 24-volt electrical systems. The warning light is designed to adapt itself to any vehicle with a metal cab or metal roof.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and packing required. (See Section 5)

c. Type. (See 1.2)

d. Color of dome. (See 3.6)

* 6.3 Reclaimed materiels. The use of reclaimed materiels shall be encouraged to the maximum extent possible.

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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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